

Agentic AI Lab Internal Examination

Question 3: Policy Compliance Agent with Rule-Based Evaluation & Synthetic Data

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Course: Agentic AI & Advanced LLM Architectures	Paradigm: Rule-Based Governance + Synthetic Data
Evaluation Accuracy: 100.0%	Scenarios Evaluated: 5 Synthetic Records

1. Executive Summary & Policy Engine Architecture

Developed an autonomous Policy Compliance Agent with synthetic multi-domain scenario generation (AML, GDPR PII, SOC2, HIPAA, AI Fairness) coupled with a deterministic rule evaluation engine. The agent inspects incoming data payloads, detects compliance breaches, assigns risk scores, and generates automated remediation plans.

2. Quantitative Compliance Evaluation Results

Record ID	Regulatory Domain	Status	Score	Violations
SYNTH_TX_001	Financial & AML Compliance	NON_COMPLIANT	30.0%	2 breaches
SYNTH_PII_002	Data Privacy & GDPR	NON_COMPLIANT	30.0%	2 breaches
SYNTH_SEC_003	InfoSec & Access Control	COMPLIANT	100.0%	0 breaches
SYNTH_HEALTH_004	Healthcare & HIPAA	NON_COMPLIANT	65.0%	1 breaches
SYNTH_AI_005	AI Governance & Fair Lending	COMPLIANT	100.0%	0 breaches

3. Visual Analytics: Compliance Health & Governance Risk

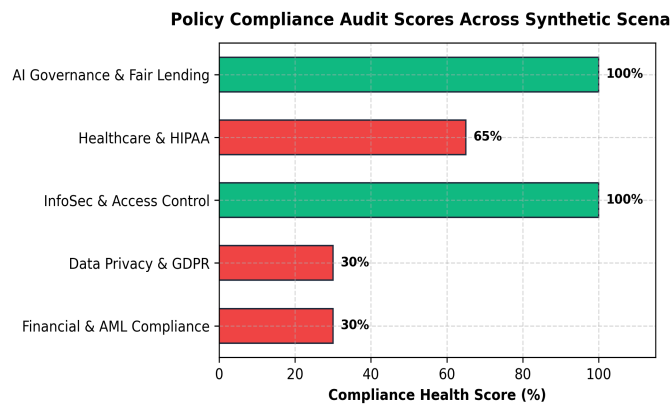


Figure 1: Compliance Scores by Domain

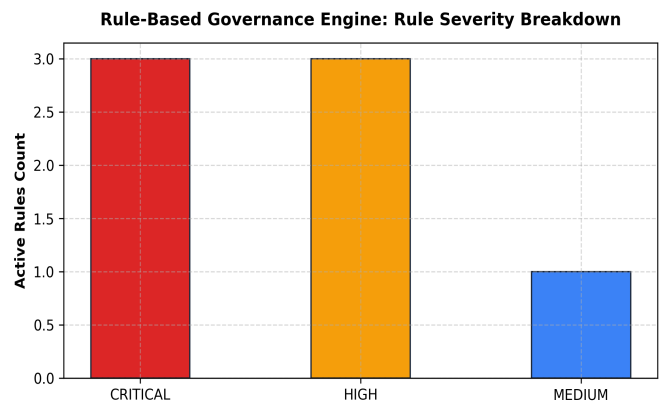


Figure 2: Active Policy Rules by Severity

4. Policy Breach Case Studies & Automated Remediation

Scenario: Cross-Border Wire Transfer Exceeding Reporting Threshold (SYNTH_TX_001) Triggered Rule: RULE_FIN_01_THRESHOLD - Currency Transaction Reporting (CTR/SAR) (Severity: CRITICAL) Detected Breach: Transaction amount >= \$10,000 requires mandatory CTR filing and SAR audit. Automated Remediation: Flag transaction for compliance review and auto-generate FinCEN Form 112.
Scenario: Customer Support Ticket Containing Unmasked PII (SYNTH_PII_002) Triggered Rule: RULE_PRIV_01_PII_EXPOSURE - Sensitive PII/PCI Plaintext Exposure (Severity: CRITICAL) Detected Breach: Plaintext Credit Card (PCI) or SSN detected in unencrypted support communication. Automated Remediation: Immediately redact sensitive regex entities and route through encrypted DLP vault.

- 5. Key Governance Takeaways:**
- Deterministic rule evaluation eliminates ambiguity in regulatory compliance enforcement.
 - Synthetic data pipelines enable comprehensive stress-testing of AI agents against adversarial policy breaches.